

Policy Implications and Future Research Directions

Implications for Policymakers and Industry Stakeholders

Our multi-level analysis yields critical insights for accelerating India's EV transition. First, policymakers should prioritize supply-side interventions over demand-side subsidies, as PLI demonstrated both immediate and sustained impacts while FAME II required extended maturation periods and proved ineffective in mature markets. Second, the extreme spatial heterogeneity (−74% to +3,808% for FAME II) necessitates abandoning one-size-fits-all approaches in favor of stage-differentiated interventions: higher subsidies for nascent markets (Assam, northeastern states), infrastructure co-investment requirements for emerging markets (Uttar Pradesh, West Bengal), and subsidy phase-outs with reallocation to charging networks in mature markets (Chandigarh, Kerala, Delhi). Third, vehicle manufacturers should coordinate product launches with infrastructure availability, as our findings show product availability was 69 percentage points more effective in high-infrastructure states, suggesting certification of minimum charging availability before entering new markets. Fourth, manufacturing-focused policies require equity adjustments to address the extreme concentration in hubs like Chhattisgarh (+327%) and Uttarakhand (+270%), through expanding PLI eligibility to component manufacturers in non-manufacturing states and establishing regional investment targets. Fifth, policy evaluation frameworks must routinely incorporate both national and state-level analysis across immediate and sustained temporal dimensions, as policies may require 12-18 months to demonstrate full impact. The comparison with China's infrastructure-first approach and decade-long policy commitment underscores that India's transition requires sustained support beyond current programs, with patience for policy maturation balanced against aggressive infrastructure expansion to maintain the current growth trajectory toward the national 30% EV penetration target by 2030.

Future Research Extensions

Future research should extend this analytical framework to examine consumer-level adoption barriers through primary survey data combined with our state-level policy effectiveness estimates, enabling identification of whether low adoption in lagging states stems from policy design failures, implementation gaps, or demand-side constraints. Additionally, a dynamic system analysis incorporating feedback loops between infrastructure deployment, vehicle availability, and adoption rates would provide more nuanced forecasting of regional convergence timelines and optimal sequencing of policy interventions across India's heterogeneous landscape.